JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SRI ANDAVAR FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

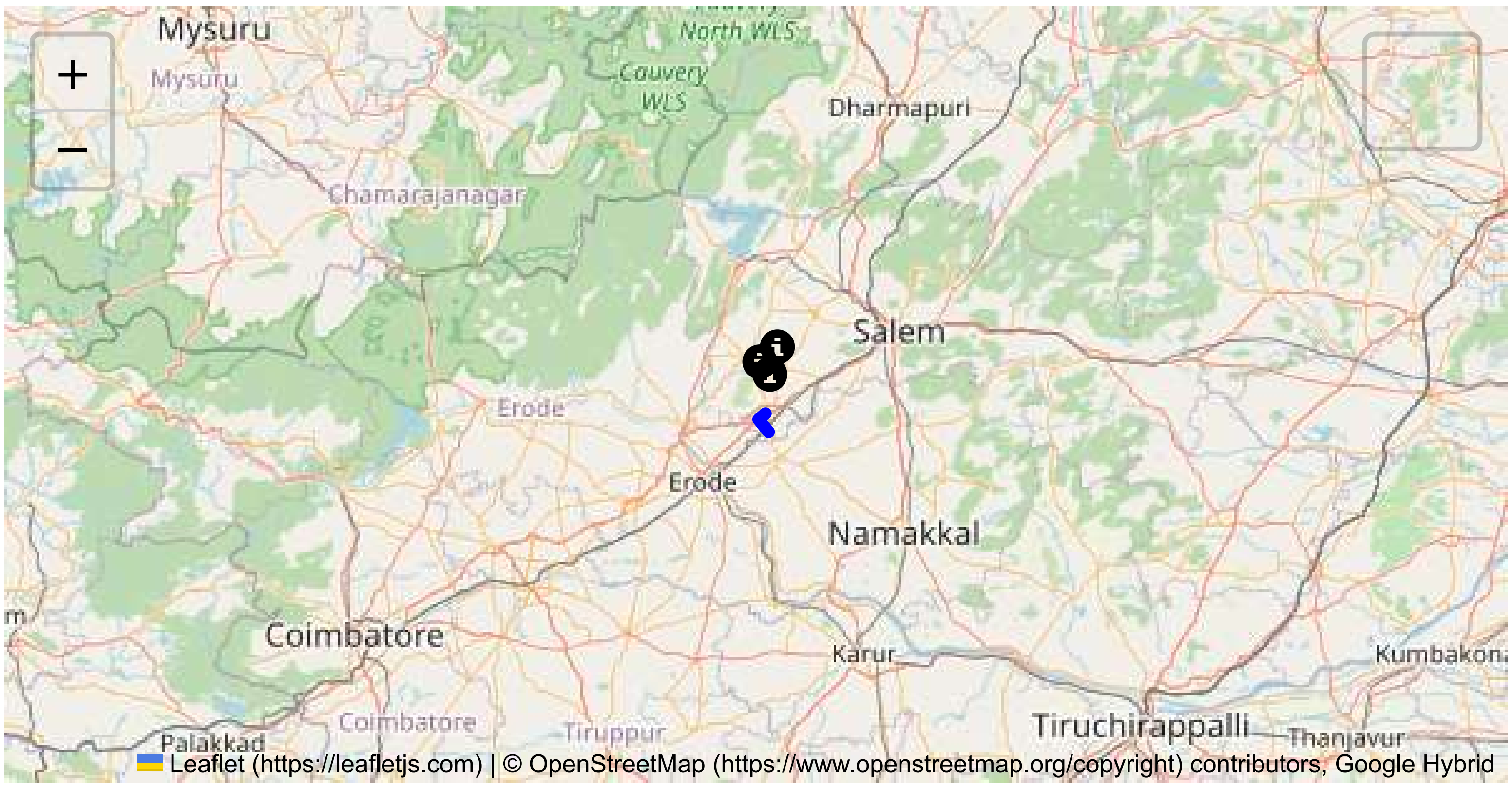
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

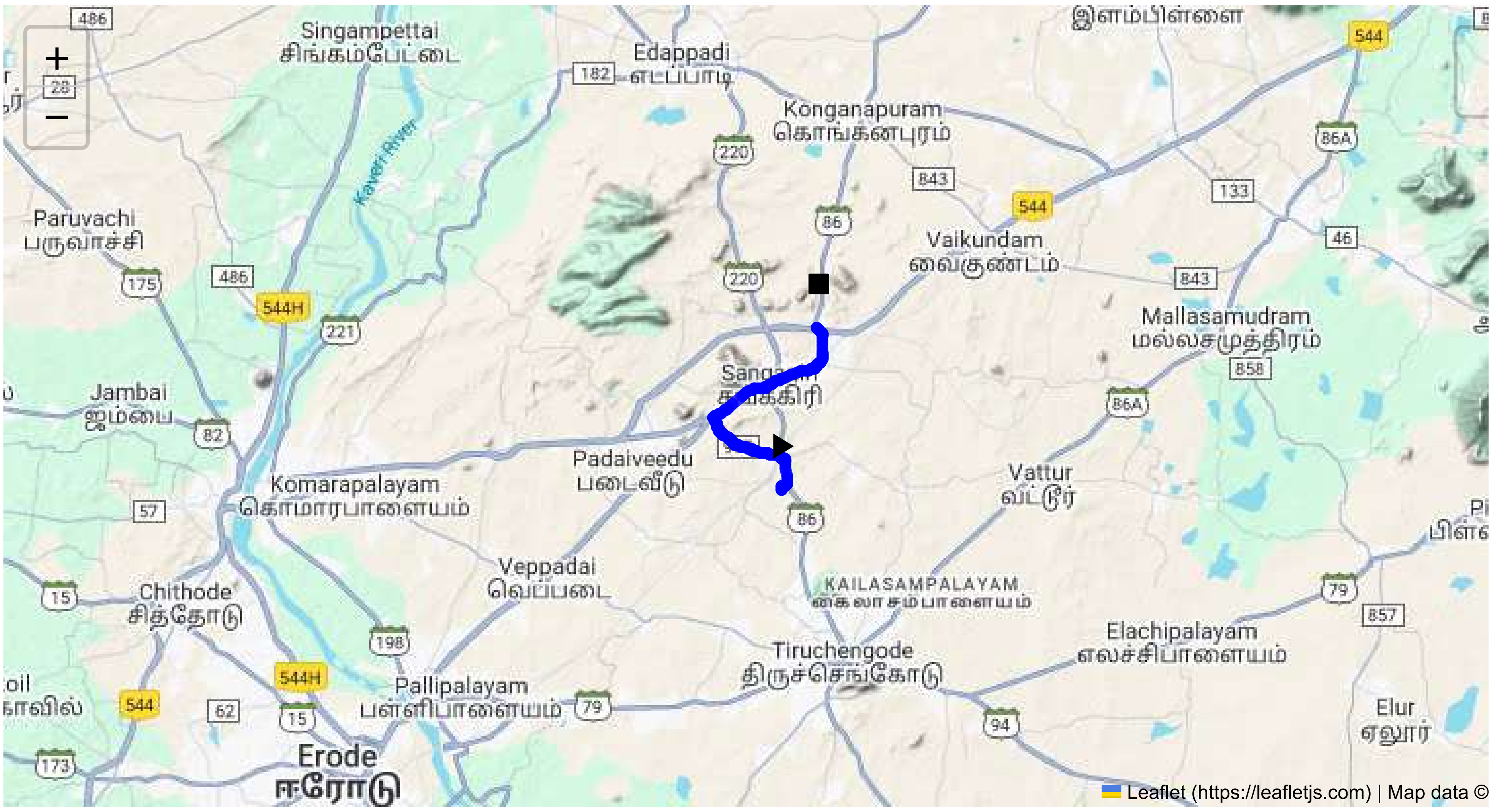
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 10.98 km
Estimated Duration: 0.3 hours
Adjusted Duration (Heavy Vehicle): 0.4 hours
Start: (11.4381, 77.8734)
End: (11.4943, 77.885254)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route spans approximately 10.98 kilometers starting from CVQF+23W in Sangagiri, Tamil Nadu, moving through FR7X+7V9, and concluding at FVVP+J73 on SH 86 in Ayveli, Tamil Nadu. This is a relatively short route, likely involving local roads and state highways that connect small towns and villages.

2. Typical Weather Conditions and Potential Weather-Related Hazards

The region experiences a tropical climate with significant rain during the monsoon months (June to September). Potential weather-related hazards include heavy rainfall leading to road flooding and reduced visibility. During the summer, high temperatures could pose risks such as equipment overheating or heat-related health concerns for drivers.

3. Analysis of Traffic Patterns

Traffic in this area is generally lighter than in urban centers. However, peak hours usually align with the standard workday schedule (8:00 AM - 10:00 AM and 5:00 PM - 7:00 PM). Congestion may occur near town centers or marketplaces, particularly in mornings and evenings when local commuters are traveling.

4. Assessment of Road Quality and Infrastructure

The road quality may vary, with potential issues including narrow lanes, potholes, and limited signage. It is crucial to be vigilant about sudden road quality changes, especially after heavy rains which can deteriorate road conditions further.

5. Suggestions for Alternative Routes for Emergencies

Alternative routes might include surrounding state highways or smaller local roads that run parallel to this main route. In case of an emergency, it is advisable to use GPS navigation systems to identify suitable detour routes that don't significantly lengthen the journey.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Tamil Nadu follows national guidelines for the transportation of hazardous materials, which require proper labeling, paperwork, and adherence to safety regulations like speed limitations and restricted transport times in populous areas. Night-time travel might be discouraged or restricted for hazardous materials.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

While specific data is not available, rural highways like SH 86 have previously seen incidents involving heavy vehicles, often due to road quality issues or driver fatigue. Ensuring that drivers are well-rested and alert is key.

8. Environmental Considerations and Sensitive Areas

Environmental sensitivities include agricultural fields and local wildlife near roadways. Drivers should be cautious to prevent contamination or accidents near these areas, particularly during the monsoon when runoff could spread pollutants.

9. Analysis of Communication Coverage

There might be sporadic communication dead zones, especially when passing through less developed areas. It is prudent to ensure that drivers are equipped with offline navigation tools and satellite communication options if available.

10. Estimated Emergency Response Times for Different Route Segments

In rural areas, emergency response times can be longer, often exceeding 30 minutes depending on the proximity to the nearest town's emergency services. Establishing contact with local authorities before transport can ensure faster response coordination.

11. Overall Summary of Risk Assessment

This route presents moderate risk with respect to road quality and narrow lanes, particularly during adverse weather conditions. Traffic congestion is minimal, but local regulations for hazardous materials require strict compliance. Alternative routes and communication strategies must be pre-planned for emergencies. Ensuring driver readiness through training can mitigate risks associated with driver fatigue and equipment failure.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43956, 77.87340	30 KM/Hr
2	Turn	Medium	11.43971, 77.87348	30 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Blind Spot	Blind Spot	11.46330, 77.84929	10 KM/Hr
5	Turn	Medium	11.48366, 77.88781	30 KM/Hr
6	Turn	Medium	11.48446, 77.88787	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	clinic	Sutha Clinic	11.473718, 77.862614	30 km/h	Medium
1	hospital	Lakshmi Hospital	11.4723746, 77.8646891	30 km/h	Medium
2	hospital	ESI Hospital	11.4717067, 77.8642668	30 km/h	Medium
3	hospital	Ck Hospital	11.473504, 77.8650523	30 km/h	Medium
4	hospital	SP Hospital	11.4708567, 77.8654872	30 km/h	Medium
5	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
6	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
7	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
8	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
9	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
11	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
13	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
14	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
10	school	Goverment Boys Higher Secondary School	11.477659, 77.8659632	30 km/h	Medium
12	marketplace	Sunday Market	11.4796502, 77.8708458	30 km/h	Medium

Route Photos of Risky Spots

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

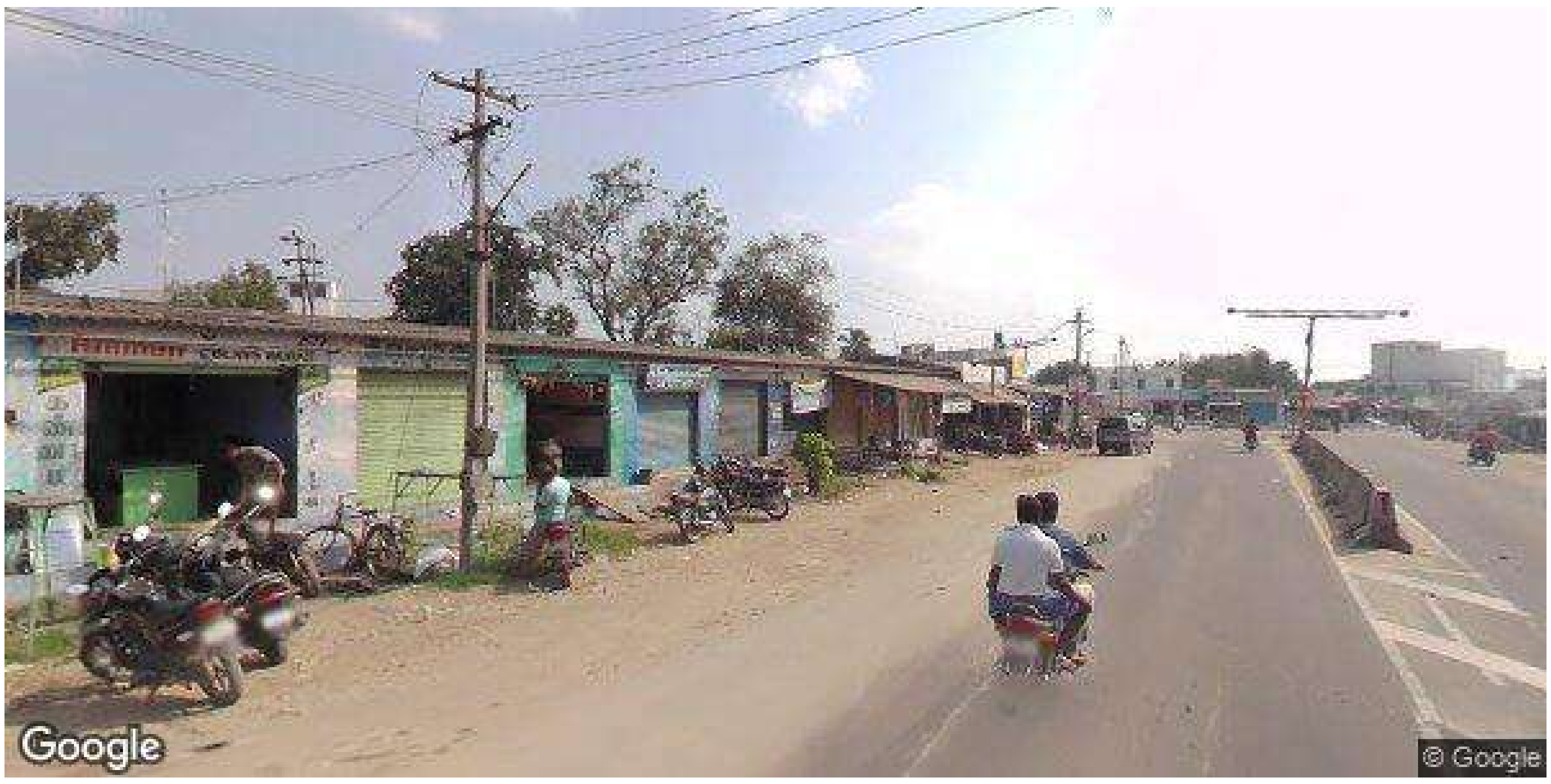
Coordinates: 11.46330, 77.84929

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.48366, 77.88781



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.48446, 77.88787